

SI 140A-02 Probability & Statistics for EECS, Fall 2026

Homework 2

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Due on Oct. 4, 2026, 23:59 UTC+8

Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.
- You are required to **submit your homework in PDF format generated by LaTeX**, otherwise you will **get no points of this homework**.

Problem 1 (25%)

A box contains 5 red balls, 4 blue balls, and 3 green balls. Three balls are drawn sequentially without replacement.

Let $A = \{\text{the first ball is red}\}$, and let $C = \{\text{at least one of the three drawn balls is red}\}$.

Suppose that the only information revealed after the experiment is that event C occurred; that is, at least one of the three drawn balls was red.

- Find $P(C)$.
- Find $P(A \cap C)$ and hence calculate $P(A | C)$.
- Let B be the event that the second ball is red. Calculate $P(B | C)$.
- Calculate $P(A \cap B | C)$.
- Compare $P(A | C)$ with $P(A)$. Are A and C independent? Justify your answer using the definition of independence.

Solution

- By the complement rule,

$$P(C) = 1 - P(\bar{C}) = 1 - \frac{\binom{7}{3}}{\binom{12}{3}} = \frac{37}{44} \approx 0.841.$$

- Since $A \subseteq C$, we have $A \cap C = A$, and

$$P(A \cap C) = P(A) = \frac{\binom{5}{1} \times 2! \times \binom{11}{2}}{\binom{12}{3} \times 3!} = \frac{5}{12} \approx 0.417.$$

Hence

$$P(A | C) = \frac{P(A \cap C)}{P(C)} = \frac{55}{111} \approx 0.495.$$

- Whether the red ball is drawn on the first attempt or the second attempt does not affect the probability. Since $B \subseteq C$, we have $P(B \cap C) = P(B) = P(A) = \frac{5}{12}$, and therefore

$$P(B | C) = \frac{P(B \cap C)}{P(C)} = \frac{55}{111} \approx 0.495.$$

- Since $A \cap B \subseteq C$,

$$P(A \cap B | C) = \frac{P(A \cap B \cap C)}{P(C)} = \frac{20}{111} \approx 0.180.$$

- By Bayes' rule with $P(C | A) = 1$,

$$P(A | C) = \frac{P(C | A)P(A)}{P(C)} = \frac{55}{111} \approx 0.495, \quad \text{while} \quad P(A) = \frac{5}{12} \approx 0.417.$$

By the definition of independence, A and C are independent if and only if $P(A \cap C) = P(A)P(C)$. Here

$$P(A \cap C) = \frac{5}{12} = \frac{220}{528} \approx 0.417, \quad P(A)P(C) = \frac{5}{12} \times \frac{37}{44} = \frac{185}{528} \approx 0.350,$$

so $P(A \cap C) \neq P(A)P(C)$, and therefore A and C are not independent.

Problem 2 (25%)

Multiple Detection System

A monitoring system uses four independent sensors to detect whether a machine has experienced a failure. Let A_i denote the event that sensor i raises a false alarm.

The false-alarm probabilities are $P(A_1) = 0.10$, $P(A_2) = 0.15$, $P(A_3) = 0.20$, $P(A_4) = 0.25$.

Assume that the four events are mutually independent.

The monitoring system issues a warning if **at least one sensor** raises a false alarm.

- Suppose the monitoring system has issued a warning. Find the probability that **at least two sensors** have raised a false alarm, i.e., $P(\text{at least two alarms} | A_1 \cup A_2 \cup A_3 \cup A_4)$.
- Suppose now that no independence assumption is made. Instead, you are given $P(A_i) = 0.1$, $i = 1, 2, 3, 4$, and $P(A_i \cap A_j) = 0.02$ for every $i \neq j$, together with $P(A_i \cap A_j \cap A_k) = 0.005$ for every distinct i, j, k , while $P(A_1 \cap A_2 \cap A_3 \cap A_4) = 0.001$.

Use the inclusion-exclusion principle to calculate $P(A_1 \cup A_2 \cup A_3 \cup A_4)$.

Solution

- Let $W = A_1 \cup A_2 \cup A_3 \cup A_4$ be the event that the system issues a warning, and let E be the event that at least two sensors raise a false alarm. Since $E \subseteq W$,

$$P(E | W) = \frac{P(E \cap W)}{P(W)} = \frac{P(E)}{P(W)}.$$

Denominator. By independence, no sensor alarms with probability

$$\begin{aligned} P(\overline{A_1} \cap \overline{A_2} \cap \overline{A_3} \cap \overline{A_4}) &= P(\overline{A_1})P(\overline{A_2})P(\overline{A_3})P(\overline{A_4}) \\ &= 0.9 \times 0.85 \times 0.8 \times 0.75 = 0.459, \end{aligned}$$

hence

$$P(W) = 1 - P(\overline{A_1} \cap \overline{A_2} \cap \overline{A_3} \cap \overline{A_4}) = 1 - 0.459 = 0.541.$$

Numerator. The event E is the union of the disjoint events “exactly k sensors alarm” for $k = 2, 3, 4$.

Exactly two alarms:

$$\begin{aligned} P(\text{exactly two}) &= 0.10 \times 0.15 \times 0.8 \times 0.75 + 0.10 \times 0.20 \times 0.85 \times 0.75 + 0.10 \times 0.25 \times 0.85 \times 0.8 \\ &\quad + 0.15 \times 0.20 \times 0.9 \times 0.75 + 0.15 \times 0.25 \times 0.9 \times 0.8 + 0.20 \times 0.25 \times 0.9 \times 0.85 \\ &= 0.009 + 0.01275 + 0.017 + 0.02025 + 0.027 + 0.03825 = 0.12425. \end{aligned}$$

Exactly three alarms:

$$\begin{aligned} P(\text{exactly three}) &= 0.10 \times 0.15 \times 0.20 \times 0.75 + 0.10 \times 0.15 \times 0.25 \times 0.8 \\ &\quad + 0.10 \times 0.20 \times 0.25 \times 0.85 + 0.15 \times 0.20 \times 0.25 \times 0.9 = 0.01625. \end{aligned}$$

Exactly four alarms:

$$P(\text{exactly four}) = 0.10 \times 0.15 \times 0.20 \times 0.25 = 0.00075.$$

Therefore

$$P(E) = 0.12425 + 0.01625 + 0.00075 = 0.14125,$$

and

$$P(E | W) = \frac{0.14125}{0.541} = \frac{565}{2164} \approx 0.261.$$

(b) By the inclusion–exclusion principle (add odd-order terms, subtract even-order terms),

$$P(A_1 \cup A_2 \cup A_3 \cup A_4) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - P(A_1 \cap A_2 \cap A_3 \cap A_4).$$

With 4 single events, $\binom{4}{2} = 6$ pairs, $\binom{4}{3} = 4$ triples and one quadruple,

$$P(A_1 \cup A_2 \cup A_3 \cup A_4) = 4(0.1) - 6(0.02) + 4(0.005) - 0.001 = 0.4 - 0.12 + 0.02 - 0.001 = 0.299.$$

Problem 3 (25%)

A factory has three production lines, L_1 , L_2 , and L_3 , which produce 30%, 45%, and 25% of the factory's total products, respectively. The probabilities that a product is defective given that it comes from L_1 , L_2 , or L_3 are $P(D | L_1) = 0.02$, $P(D | L_2) = 0.03$, $P(D | L_3) = 0.05$, where D denotes the event that a randomly selected product is defective.

- Two products are selected independently from the factory. Find the probability that both products are defective.
- Three products are selected independently from the factory. Find the probability that at least one of the three products is defective.
- Suppose n products are selected independently from the factory. Find the smallest integer n such that the probability of obtaining at least one defective product is at least 95%.

Solution

- By the law of total probability,

$$\begin{aligned} P(D) &= P(L_1)P(D | L_1) + P(L_2)P(D | L_2) + P(L_3)P(D | L_3) \\ &= 0.30 \times 0.02 + 0.45 \times 0.03 + 0.25 \times 0.05 = 0.032. \end{aligned}$$

Since the two products are selected independently,

$$P(\text{both defective}) = P(D)^2 = 0.032^2 \approx 0.00102.$$

- By the complement rule, the probability that none of the three products is defective is

$$(1 - P(D))^3 = 0.968^3 \approx 0.907,$$

hence

$$P(\text{at least one defective}) = 1 - 0.968^3 \approx 0.0930.$$

- For n selected products,

$$P(\text{at least one defective}) = 1 - (1 - P(D))^n = 1 - 0.968^n \geq 0.95,$$

which is equivalent to $0.968^n \leq 0.05$. Taking logarithms,

$$n \geq \frac{\ln 0.05}{\ln 0.968} \approx 92.1.$$

Indeed $0.968^{92} \approx 0.0502 > 0.05$ while $0.968^{93} \approx 0.0485 \leq 0.05$, so the smallest such integer is $n = 93$.

The numerical values above are computed with Python; the code `hm2.py` and its output are shown below.

```
Sep 28 21:10 Mon
12 p = 0.38 * 0.02 + 0.45 * 0.03 + 0.25 * 0.05
11 q = 1 - p
10
9 n = 0
8 prob = 0.0
7 while prob < 0.95:
6     n += 1
5     prob = 1 - q ** n
4
3 print(f"P(D) = {p}")
2 print(f"P(both defective) = {p ** 2}")
1 print(f"P(at least one in 3) = {1 - q ** 3}")
13 print(f"smallest n = {n}, P(at least one) = {prob}")

P(D) = 0.032
P(both defective) = 0.001024
P(at least one in 3) = 0.092960768000000008
smallest n = 93, P(at least one) = 0.9514255122157417
[Process exited 0]
```

Problem 4 (25%)

Two dice are rolled independently. For each die, the probability of showing an even number is p , where $0 < p < 1$. The two dice have the same distribution, but no assumption is made that the individual outcomes $1, 2, \dots, 6$ are equally likely.

Define $A = \{\text{the sum of the two dice is even}\}$, and $B = \{\text{the sum of the two dice is greater than 8}\}$.

Determine whether there exists a value of p such that, for every possible distribution of the individual die outcomes satisfying $P(\text{even}) = p$, the events A and B are independent.

If such a value of p exists, find all such values and prove that the independence holds for every admissible distribution.

If no such value exists, prove this by constructing, for each candidate value of p , two admissible distributions that have the same value of p but give different values of $P(A \cap B) - P(A)P(B)$.

Your argument should distinguish carefully between independence that is implied by the given information and independence that occurs only for a particular choice of the underlying distribution.

Solution

No such value of p exists.

Note first that $P(A)$ depends on p only: the two dice are independent and identically distributed, so the sum is even exactly when both outcomes are even or both are odd, hence

$$P(A) = p^2 + (1 - p)^2.$$

The quantities $P(B)$ and $P(A \cap B)$, however, depend on how the mass is placed on the faces $1, \dots, 6$ and are not determined by p .

Fix any $p \in (0, 1)$ and take the admissible distribution

$$P(X = 3) = 1 - p, \quad P(X = 6) = p.$$

Then $X + Y \in \{6, 9, 12\}$ with

$$P(X + Y = 6) = (1 - p)^2, \quad P(X + Y = 9) = 2p(1 - p), \quad P(X + Y = 12) = p^2.$$

Since only 9 and 12 exceed 8,

$$P(B) = P(X + Y > 8) = 2p(1 - p) + p^2,$$

and since 12 is even,

$$P(A \cap B) = P(X + Y = 12) = p^2.$$

Therefore

$$\begin{aligned} P(A \cap B) - P(A)P(B) &= p^2 - (p^2 + (1 - p)^2)(2p(1 - p) + p^2) \\ &= -2p(1 - p)^3 \neq 0 \quad \text{for every } 0 < p < 1, \end{aligned}$$

so under this distribution A and B are not independent.

For the same p , the distribution $P(X = 5) = 1 - p$, $P(X = 6) = p$ is also admissible and gives $P(B) = 1$ together with $P(A \cap B) = P(A)$, hence

$$P(A \cap B) - P(A)P(B) = 0.$$

Two admissible distributions sharing the same p therefore produce different values of $P(A \cap B) - P(A)P(B)$, so no single p makes A and B independent for every admissible distribution.

Thus $P(A) = p^2 + (1 - p)^2$ is the only thing forced by the constraint $P(\text{even}) = p$; independence would be a property of particular distributions and is never forced by that constraint.